

Kunal Pai

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EDUCATION

Ph.D., Computer Science, University of California, Los Angeles Enrolled: September 2026
M.S., Computer Science, University of California, Davis (GPA: **4.0/4.0**) June 2026
B.S., Computer Science & Engineering, University of California, Davis (GPA: **3.8/4.0**), with Honors June 2023

RELEVANT SKILLS

Languages: Python, C++, C, Rust, JavaScript, TypeScript, Java, Bash

ML/AI: LLMs, Multi-Agent Systems, AI Agent Security, Prompt Engineering, PyTorch, TensorFlow, Hugging Face, Ollama, scikit-learn, Model Calibration, Evolutionary Algorithms

Systems & Compilers: gem5, Computer Architecture Simulation, RISC-V, LLVM, Clang, C-to-Rust Transpilation, Hardware Profiling (perf)

Web/Data: React, Next.js, Django, Flask, MongoDB, pandas, NumPy, Matplotlib, REST APIs

Tools: Git, Docker, Unix/Linux, Jupyter, LaTeX, Slurm / HPC, GitHub Actions

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Student Researcher, DavSec Lab @ UC Davis

April 2025 – June 2026

Research Project

Python, LLMs, C++, Rust, Compiler Analysis

- Built automated C-to-Rust transpilation pipeline using LLMs with 5 prompt variations, benchmarking state-of-the-art models across 746 programs and achieving 70.2% functional accuracy
- Engineered a multi-variant performance-aware transpilation framework that explores alternative Rust implementations in an evolutionary agent setting, yielding up to 3× runtime reduction across benchmarks
- Developed cross-layer analysis framework combining compiler metrics with hardware performance counters to expose semantic and optimization gaps in LLM-translated, syntax-directed and manually-written Rust translations

Graduate Student Researcher, DECAL Lab @ UC Davis

September 2022 – December 2024

Research Project

Python, LLMs, Machine Learning, Code Analysis

- Developed a 4,500-sample dataset for pairwise code-documentation alignment from 200 open-source Python projects, enabling future research in software maintenance
- Engineered a pipeline for measuring calibration and correctness of large language models for code repair, using Defects4J
- Quantified precision and recall gains (ROUGE, METEOR) from semantic augmentation in LLM prompt pipelines for code summarization

Teaching Assistant, University of California, Davis

September 2023 – December 2023

- Assisted 180 students in a senior-level Probability & Statistical Modeling class.

PUBLICATIONS (SELECTED)

Kiwi: Slicing Models and Reconciling Adapters in Fault-Tolerant, Decentralized LLM Fine-Tuning, Patel, H., Shah, P., Shah, S. & Pai, K., CODEC-FM 2026: Collaborative, Open, and Decentralized Foundation Models Workshop, NeurIPS 2026

VATS: Exploiting Implicit Authority in Error-Path Injection via Systematic Mutation [SPOTLIGHT], Patel, H. & Pai, K., Second Workshop on Agents in the Wild: Safety, Security, and Beyond, ICML 2026

Toward Reproducible and Standardized Computer Architecture Simulation with gem5, Pai, K., Patel, H., Le, E., et. al., IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS) 2026

NAAMSE: Framework for Evolutionary Security Evaluation of Agents, Pai, K., Shah, P. & Patel, H., ICLR 2026 Agents in the Wild: Safety, Security, and Beyond Workshop

Implications of Full-System Modeling for Superconducting Architectures, Pai, K., Samani, M., Nand, A. & Lowe-Power, J., Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis (SC Workshops '25)

CoDocBench: A Dataset for Code-Documentation Alignment in Software Maintenance, Pai, K., Devanbu, P. & Ahmed, T., International Conference on Mining Software Repositories (MSR) 2025: Data and Tool Showcase Track

Calibration and Correctness of Language Models for Code, Spiess, C., Gros, D., Pai, K.S., et. al., International Conference on Software Engineering (ICSE) 2025

Automatic semantic augmentation of language model prompts (for code summarization), Ahmed, T., Pai, K.S., Devanbu, P. & Barr, E.T., International Conference on Software Engineering (ICSE) 2024

PROJECT EXPERIENCE

NAAMSE: Neural Adversarial Agent Mutation-based Security Evaluator <i>Research Project</i>	Nov 2025 – Present <i>Python, LLMs, Evolutionary Algorithms</i>
• Won 2nd place (Agent Safety) at UC Berkeley RDI AgentBeats Competition; framework infrastructure was subsequently forked by Mozilla's Odin team • Designed and implemented a clustering engine to identify semantic attack vectors in LLM-generated adversarial prompts • Engineered an attack pipeline that iteratively generates, evaluates, and refines adversarial prompts against target models • Benchmarked multiple frontier LLMs within the framework to validate attack effectiveness and refine scoring metrics	
HASHIRU: Hierarchical, Resource-Aware Multi-Agent Framework <i>Research Project</i>	March 2025 – June 2025 <i>Python, LLMs, Multi-Agent Systems</i>
• Designed and deployed a multi-agent architecture enabling dynamic, LLM-driven collaboration across diverse tasks • Implemented task decomposition with intelligent agent delegation based on resource cost models and task specialization • Engineered autonomous generation of tools and APIs for task execution • Developed a robust evaluation framework for agent performance across complex, multi-step tasks	
SuperNOVA: Superconducting Graph Accelerator in gem5 <i>Research Project, DArchR Lab @ UC Davis</i>	January 2024 – March 2026 <i>C++, Python, gem5, Hardware Simulation</i>
• Modeled a superconducting graph accelerator and interconnect in gem5, achieving up to 24× speedup and 73× energy efficiency • Integrated cryogenic/superconducting cores, caches, and interconnects to evaluate bottlenecks for realistic workloads • Mentored undergraduates on modeling, benchmarking, and research writing; one co-authored a ModSim 2024 poster	
gem5 Vision <i>Resource Discovery Framework</i>	January 2023 – June 2023 <i>Next.js, Python, MongoDB, JSON Schema</i>
• Accelerated resource discovery 20× for 1,200+ gem5 artifacts by optimizing search and categorization logic • Implemented categorization and semantic versioning across 20+ resource types to streamline retrieval • Integrated local and remote JSON schemas with MongoDB, improving accessibility for 500+ users	

AWARDS AND HONORS

Distinguished Artifact Reviewer Award , ISSTA 2026 Artifact Evaluation	2026
Finalist (Agentic Financial & High-Stakes AI Advice) , Cambridge C:\>DIR Global 'Agentic Regulator' Hackathon	2026
2nd Place (Agent Safety) , UC Berkeley RDI AgentBeats Competition	2026
Dean's List , UC Davis College of Engineering	Fall 2019, Fall 2020, Winter 2022, Spring 2022
Provost Award , UC Davis	2019–2023

SERVICE

Program Committee
• MSR, Data and Tool Showcase Track (2026, 2027)
Artifact Evaluation Committee
• ISSTA (2026 – Distinguished Reviewer) • ASE (2026) • ISPASS (2026)
Reviewer
• NeurIPS, CODEC-FM Workshop (2026) • NeurIPS, SocialAgent Workshop (2026) • NeurIPS, Agents in the Wild Workshop (2026) • ICML, Agents in the Wild Workshop (2026)